L05: Rank of the "Data Matrix"

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Recap

1. Representation as a vector in R^d
   - All: Web page; Image; Song; Weather.
   - If we have N samples we can have a $N \times d$ Data Matrix $\mathbf{X}$.
   - $\mathbf{X}^T\mathbf{X}$ is a scaled covariance matrix Σ, if data is mean subtracted.
   - What is rank of $\mathbf{X}$? min (N,d). !!
2. Learn a function $y = f(\mathbf{W}, \mathbf{x})$ from the data
   - Notion of Training, Validation and Testing; Loss Function
   - Role of Optimization: (i) GD (ii) Closed form (iii) EVec
3. Feature Transformation as a useful trick:
   - $\mathbf{x}' = f(\mathbf{W}, \mathbf{x})$; $\mathbf{x}' = \mathbf{W}\mathbf{x}$ and Dimensionality Reduction
4. Classification and Regression Algorithms:
   - Nearest Neighbour: Decide based on majority labels of K NNs
   - Linear Classification: Decide as ω_1 if $\mathbf{w}^T x \geq 0$ Else ω_2
   - Decide as ω_1 if $P(\omega_1|\mathbf{x}) \geq P(\omega_2|\mathbf{x})$ else ω_2
   - Linear, Ridge and Lasso Regression
   - Performance Metrices
5. Low Rank Data MatriX
   - Eigen Values, Eigen Vectors and SVD
   - Factorization and Low Rank approximation

Revise

Revise your understanding on the following topics:

- Matrices and Properties
- PSD and Types of Matrices
- Eigen values and Eigen vectors
- Determinants
- Rank
- SVD and Matrix Decompositions/Factorizations

Book: `https://mml-book.github.io/` Most of Chapter 2 and Chapter 4.

Bag of Words Histogram and One-Hot Representation

The Bag of Words Representation

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!

it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
great	1
...	...

A simple exercise

Many years back, Someone took all the speeches of past US presidents and created a Bag of Words Histograms. (one histogram per President). You can visualize the histogram as a word cloud (see next slide). Higher the frequency, the larger the font.

Q:Who are the presidents in the next three slides? When were they presidents (year?)

Q: How will the histograms be different, if it was done for each speech separately?

Q: Guess what it could be the same for Joe Baiden and Donald Trump. Try out yourself after the class.!!

Representation: Bag of Words and One-Hot

Representation: Bag of Words and One-Hot

Representation: Bag of Words and One-Hot

Discussion Point

We consider 100 documents each from "sports" "politics" and "finance" and create a BoW representation of size 300.

1. We construct a Data Matrix D by keeping each vector as a row. What is the dimension of this matrix? (simple!). Q: Any need to normalize this histogram?
2. Q: What could be the rank of this matrix? (wait!!. let us answer the next two questions first.)
3. Assume there was a small error in the code in that created this matrix. i.e., a single sports document was copied 100 times instead of different 100 articles (and similarly for polictics and finance). What will be the rank of the data matrix?
4. Assume a different situation. It was only one sports article; and it was "rewritten" by 100 journalists (like plagiarism (i) by rearranging words (ii) by minorly rewriting). What will be the rank of the data matrix?

Discussion Point: Binary Class

- A TA wanted to create a simple two class dataset for an assignment. TA started with two very distinct d vectors and added some small noise (along all dimensions). Finally got, N samples in two well separated classes. What is the rank of the Data matrix?
- How will the eigen value spectrum look like?
- What is the rank of a data matrix $\mathbf{X}$ of N samples, from p concepts, and each sample is of d dimension.

Discussion Point

Without numerically computing, what is the rank of the following matrix?

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

1. 1
2. 2
3. 3
4. 4

Hint:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 1+3 & 2+3 & 3+3 \\ 1+6 & 2+6 & 3+6 \end{bmatrix}$$

Discussion Point

Consider a matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} + \epsilon B$$

where B is a full rank matrix (say I or formed by random integers in 1 to 100).

- When $\epsilon = 1$, what is the rank of A?
- When $\epsilon = 10^{-100}$, what is the rank of A?
- How many "non-zero" eigen values will A have? Can you comment on their magnitude in both cases?

Problem

We know that the rank of a 3×3 matrix formed by first 9 numbers arranged sequentially is 2.

What is the rank of a 5×5 matrix formed by first 25 numbers arranged sequentially?

Eigen Values and Vectors

- $\mathbf{Av} = \lambda\mathbf{v}$, where $\mathbf{A}$ is a $N \times N$ matrix
- $\lambda_1, \ldots, \lambda_N$ are tyhe eigen values
- $\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_N}$ are the eigen vectors.
- Eigen vectors are orthogonal $\mathbf{v_i^T v_j} = 0$
- Product of eigern values is determinant.
- Number of non-zero eigen values is the rank.
- $\mathbf{A}$ is PSD iff $\mathbf{x}^T\mathbf{Ax} \geq 0 \forall \mathbf{x}$

Eigen Decomposition

Popular eigen expansion

$$A = \sum_{i=1}^{d} \lambda_i \mathbf{v}_i \mathbf{v}_i^T$$

Also

$$A = U \Lambda U^T$$

is known as the eigen decomposition.

SVD

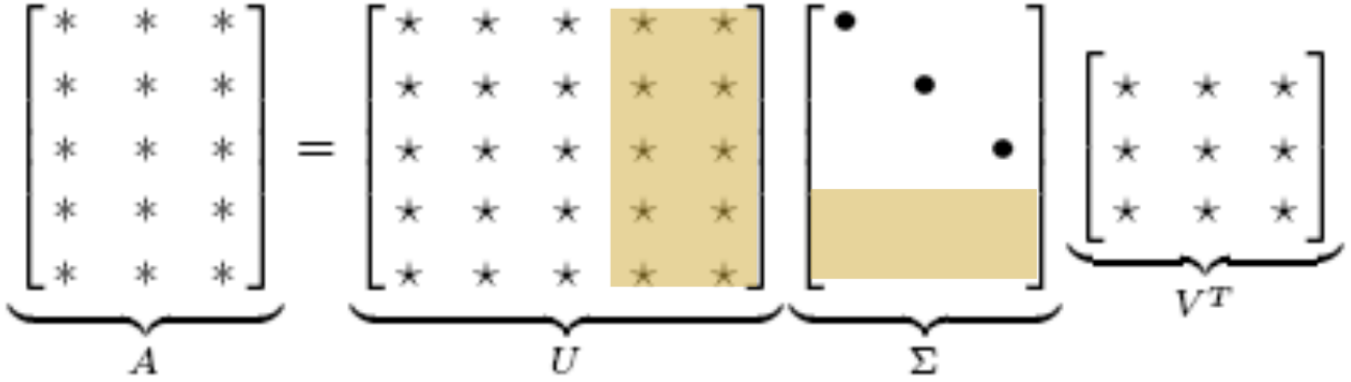
Formally, the **singular value decomposition** (SVD) of an $m \times n$ real or complex matrix M is a factorization of the form

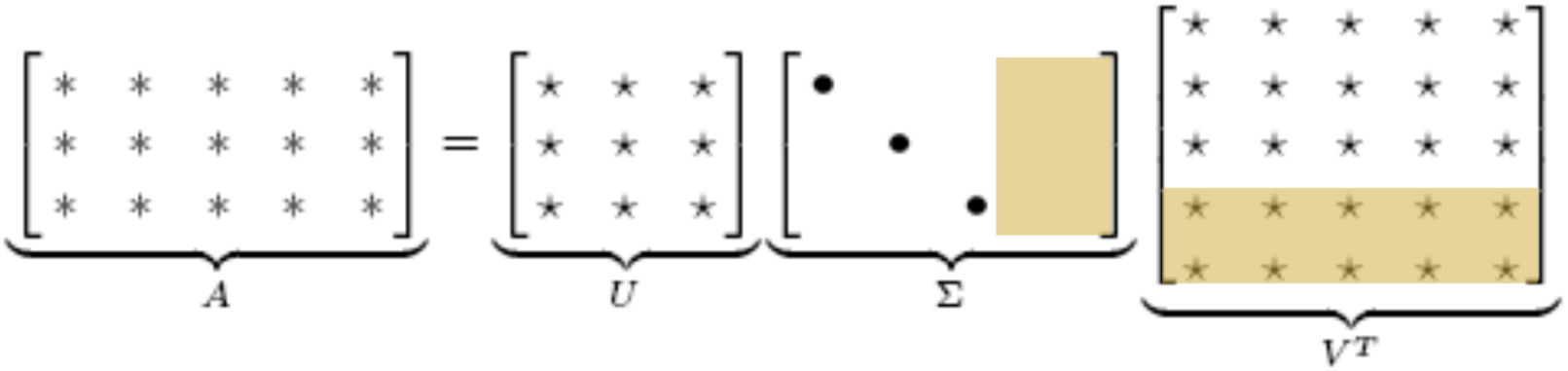
$$M = UDV^T$$

- U is $m \times n$, $U^T U = I_n$,
- D is diagonal $n \times n$ and
- V is $V^T V = VV^T = I_n$.

$$M = \sum_i D_i u_i v_i^T$$

SVD





Numerical Example

Example 1

$$A = \begin{bmatrix} \mathbf{3} & \mathbf{0} \\ \mathbf{4} & \mathbf{5} \end{bmatrix}$$

$$\boldsymbol{U} = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix} \quad \boldsymbol{\Sigma} = \begin{bmatrix} \sqrt{45} & \\ & \sqrt{5} \end{bmatrix} \quad \boldsymbol{V} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}.$$

Example 2

$$\begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix} = \begin{bmatrix} 1/3 & 2/\sqrt{5} & 2/\sqrt{5} \\ -2/3 & 1/\sqrt{5} & 0 \\ -2/3 & 0 & 1/\sqrt{5} \end{bmatrix} \begin{bmatrix} 3\sqrt{10} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -3/\sqrt{10} & 1/\sqrt{10} \\ 1/\sqrt{10} & 3/\sqrt{10} \end{bmatrix}^T$$

$$\begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix} = \begin{bmatrix} 1/3 \\ -2/3 \\ -2/3 \end{bmatrix} 3\sqrt{10} \begin{bmatrix} -3/\sqrt{10} \\ 1/\sqrt{10} \end{bmatrix}^T = \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix} [-3, 1]$$

$$\sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^{\mathrm{T}} + \sigma_2 \boldsymbol{u}_2 \boldsymbol{v}_2^{\mathrm{T}} = \frac{\sqrt{\mathbf{45}}}{\sqrt{\mathbf{20}}} \begin{bmatrix} \mathbf{1} & \mathbf{1} \\ \mathbf{3} & \mathbf{3} \end{bmatrix} + \frac{\sqrt{\mathbf{5}}}{\sqrt{\mathbf{20}}} \begin{bmatrix} \mathbf{3} & \mathbf{-3} \\ \mathbf{-1} & \mathbf{1} \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} = A.$$

Eigen vs SVD

We know that:

- Eigen Decomposition of Symmetric Matrix **S**

$$\mathbf{S} = \mathbf{Q}\Lambda\mathbf{Q}^T = \sum_{i=1}^{n} \lambda_i \mathbf{q}_i \mathbf{q}_i^T$$

- SVD of **A**

$$\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^T = \sum_{i=1}^{n} d_i \mathbf{u}_i \mathbf{v}_i^T$$

- The non-zero singular values of M (found on the diagonal entries of D) are the square roots of the non-zero eigenvalues of both $M^T M$ and MM^T.

Problem 1

- Consider the $N \times d$ data matrix $\mathbf{X}$ with every column as data elements. Show that $\mathbf{A} = \mathbf{X}^T\mathbf{X}$ is Symmetric. Show that $\mathbf{A}$ is PSD. [1]
- Consider a data matrix $\mathbf{X}$ where every column is mean subtracted samples. We compute $\mathbf{A} = \mathbf{X}^\mathbf{T}\mathbf{X}$. Then $\mathbf{A}$ (agree/disagree)
  - is a $d \times d$ matrix
  - is a $N \times N$ matrix
  - is a symmetric matrix
  - is a scaled version of the covariance matrix

[1] A matrix A is PSD if $z^T A z \geq 0 \ \forall z \in R^d$.

Problem 2

We know that:

- Eigen Decomposition of Symmetric Matrix $\mathbf{S}$

$$\mathbf{S} = \mathbf{Q}\Lambda\mathbf{Q}^T = \sum_{i=1}^{n} \lambda_i \mathbf{q}_i \mathbf{q}_i^T$$

- SVD of $\mathbf{A}$

$$\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^T = \sum_{i=1}^{n} d_i \mathbf{u}_i \mathbf{v}_i^T$$

1. How do we compute $\mathbf{S}^{-1}$ and $\mathbf{A}^{-1}$
2. If $\mathbf{A} = \mathbf{p}\mathbf{q}^T$ is a 3×3 matrix, what is the SVD of $\mathbf{A}$?

Low Rank Approximation

Problem: Given a $N \times N$ matrix $\mathbf{A}$, we can find a closest low rank, (say p) (where $p < N$), matrix A'.

$$\mathbf{A}' = \arg\min_{\mathbf{M}} ||\mathbf{A} - \mathbf{M}|| \text{ such that rank of } \mathbf{M} \text{ is p}$$

Algorithm:

1. Compute SVD of $\mathbf{A} \rightarrow UDV^T$
2. Make all singular values beyond p zero. (i.e., $D_{jj} \leftarrow 0$ for $j > p$)
3. Reconstruct back as $\mathbf{A}' \leftarrow UDV^T$.

Review Question - I (one, none or more correct)

Confusion matrix is:

(a) Square (b) Always Diagonal (c) Can never be diagonal (d) Can be diagonal (e) Always Symmetric (f) Can never be symmetric (g) Can be Symmetric

Review Question - II (one, none or more correct)

Covariance Matrix:

- (a) Can neve be diagonal.
- (b) Can be diagonal
- (c) Always Positive Semi Definite
- (d) Always full rank
- (e) Never full rank.
- (f) Always Symmetric
- (g) Not guaranteed to be symmetric

Review Question - III (one, none or more correct)

Let $\mu = \sum_{i=1}^{N} \mathbf{x}_i$ be the mean of $\mathbf{x}_1, \ldots, \mathbf{x}_N$. $\mathbf{x}_i \in R^d$

- (a) μ is also $\in R^d$
- (b) μ is always one of the N samples.
- (c) μ can not be one of the N samples
- (d) μ can be one of the N samples.
- (e) μ is equidistant from all the N samples
- (f) μ has the least sum of square error from all the samples (i.e., μ is $\arg\min_{\mathbf{y}} \sum_{i=1}^{N} ||\mathbf{x}_i - \mathbf{y}||_2^2$)